

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083)

TIME ALLOWED : 1 Hr. 45 Min.

General Instructions:

1. The paper is divided into 5 Sections - A, B, C, D and E.
2. All questions are compulsory.
3. All programming questions are to be answered using Python Language only.
4. In case of MCQs, text of the correct answer should also be written.
5. Write all the sub-parts of a question together sequentially.

SECTION - A

Q.1 Answer the following: (1 X 13 = 13)

(a) State True or False:

"A user-defined function in Python can return more than one value at a time."

(b) What will be the output of the following code?

```
def Update(A):  
    A = A * 2  
    print(A, end='#')  
A = 7  
Update(A)  
print(A)
```

(A) 14#7

(B) 7#14

(C) 14#14

(D) 7#7

(c) Which of the following is a valid function header?

(A) def calc(a=5, b):

(B) def calc(a, b=5):

(C) def calc(a=5, b, c=2):

(D) def calc(a, b=5, c):

(d) Which Python module must be imported to use the `dump()` and `load()` methods on a binary file?

(A) csv

(B) pickle

(C) os

(D) binary

(e) Which file mode opens an existing binary file for both reading and writing without erasing its contents?

(A) 'wb+'

(B) 'rb+'

(C) 'ab'

(D) 'w+'

(f) State True or False:

"An absolute path of a file always starts from the root directory (or drive) of the system."

(g) Write a statement to place the file pointer of the file object `F` at the 20th byte from the beginning of the file.

- (h) Which of the following is not a guided (wired) transmission medium?
- (A) Twisted pair cable (B) Co-axial cable
(C) Microwave (D) Fibre-optic cable
- (i) _____ protocol allows a user to log in to a remote computer and use its resources.
- (A) PPP (B) TELNET
(C) SMTP (D) HTTP
- (j) In which network topology does the failure of the single backbone cable bring down the entire network?
- (A) Star (B) Bus
(C) Tree (D) None of these
- (k) Data transfer rate of a communication medium is measured in:
- (A) Hertz (Hz) (B) bits per second (bps)
(C) metres (D) decibels
- Q1 (l) and (m) are Assertion (A) and Reason (R) based questions. Mark the correct choice as:
- (A) Both A and R are True and R is the correct explanation for A.
(B) Both A and R are True and R is not the correct explanation for A.
(C) A is True but R is False.
(D) A is False but R is True.
- (l) **Assertion (A):** A variable created inside a function cannot be accessed outside that function.
Reason (R): The scope of a local variable is limited to the function in which it is defined.
- (m) **Assertion (A):** HTTPS is considered safer than HTTP for online banking.
Reason (R): HTTPS encrypts the data exchanged between the web browser and the web server.

SECTION - B

- Q.2** (a) Differentiate between a Hub and a Switch. (1+1)
(b) Expand the following: PPP, POP3
- Q.3** What will be the output of the following Python code? (2)
- ```
def modify(L, x):
 L.append(x)
 x = x * 2
 L[0] = L[0] + x
 return x

nums = [5, 10]
val = 3
r = modify(nums, val)
print(nums, val, r)
```
- Q.4** Differentiate between the `write()` and `writelines()` methods of a file object. Give one example of each. (2)

### SECTION - C

**Q.5** Write a function **COUNT\_CHARS()** in Python that reads the text file "**DATA.TXT**" and displays the number of uppercase letters, lowercase letters and digits present in the file. (3)  
For example, if the file contains: CBSE Exam 2027 is on 15th March.  
the output should be: Uppercase: 6    Lowercase: 13    Digits: 6

**Q.6** A text file "**POEM.TXT**" contains the following lines: (3)

```
Twinkle twinkle little star
How I wonder what you are
Up above the world so high
Like a diamond in the sky
```

Write the output of each of the following code segments:

```
(a) f = open("POEM.TXT", "r")
 print(f.read(7))
 f.close()
```

```
(b) f = open("POEM.TXT", "r")
 L = f.readlines()
 print(len(L), L[2].split()[0])
 f.close()
```

```
(c) f = open("POEM.TXT", "r")
 f.seek(8)
 print(f.read(7))
 print(f.tell())
 f.close()
```

**Q.7** Write a Python function **FREQ(L)** that accepts a list **L** of integers as argument and returns a dictionary in which each distinct element of the list is a key and its number of occurrences is the value. (3)

For example, if **L = [4, 2, 4, 7, 2, 4]**, the function should return **{4: 3, 2: 2, 7: 1}**.

Also write a statement to call the function and display the returned dictionary.

### SECTION - D

**Q.8** Ananya is maintaining the records of a school library in a CSV file named "**LIBRARY.CSV**". Each record has the fields [**BookID**, **Title**, **Author**, **Qty**]. Help her by writing the following user-defined functions: (4)

(a) **AddBook()** – to accept the details of a book from the user and append it as a record to "**LIBRARY.CSV**".

(b) **ShowLowStock()** – to read "**LIBRARY.CSV**" and display the details of all books whose Qty is less than 5, along with the total number of such books.

**Q.9** Answer the following questions: (4)

(a) What was ARPANET? Name the network that evolved from it in the mid-1980s for academic research.

(b) Write one difference between Circuit Switching and Packet Switching.

(c) Why is microwave transmission called line-of-sight transmission?

(d) What is a domain name? Identify the domain name in the URL:  
`https://www.cbse.gov.in/web/index.html`

**Q.10** Write the output of the following code segments:

**(2+2)**

```
(a) p = 5
 def calc(q):
 global p
 p = p + q
 q = q + p
 return q
 r = calc(p)
 print(p, r)
 r = calc(r)
 print(p, r)

(b) def show(name, marks=40, grade='C'):
 if marks >= 80:
 grade = 'A'
 print(name, marks, grade, sep='-')
 show('Riya', 85)
 show('Amit', grade='B')
 show(marks=90, name='Joy', grade='B')
 show('Sam', 60, 'B')
```

### **SECTION - E**

**Q.11** A binary file "**STUDENT.DAT**" stores records of students, where each record is a list in the form [AdmNo, Name, Percent]. **(1+2+2)**

(a) Differentiate between the 'wb' and 'ab' file modes.

(b) Write a function **addrec()** to accept the details of a student from the user and append the record to the file "STUDENT.DAT".

(c) Write a function **update\_percent(adm)** that accepts an admission number as argument and increases the Percent of that student by 5 in the file "STUDENT.DAT". If no such student exists, display "Record not found".

- Q.12 Global Tech Solutions** is setting up a new campus in Durgapur with four blocks – ADMIN, TECH, HR and ACCOUNTS. Its head office is in Delhi. As a network expert, you have to suggest the best network solutions based on the following details: **(5)**

| Blocks            | Distance |
|-------------------|----------|
| ADMIN to TECH     | 60 m     |
| ADMIN to HR       | 120 m    |
| ADMIN to ACCOUNTS | 90 m     |
| TECH to HR        | 50 m     |
| TECH to ACCOUNTS  | 110 m    |
| HR to ACCOUNTS    | 70 m     |

| Block    | No. of Computers |
|----------|------------------|
| ADMIN    | 25               |
| TECH     | 120              |
| HR       | 30               |
| ACCOUNTS | 20               |

- (a) Suggest the most suitable block to house the server of this organisation. Justify your answer.
- (b) Suggest and draw a cable layout to efficiently connect the blocks within the Durgapur campus.
- (c) Suggest the placement of the following devices: (i) Repeater (ii) Switch
- (d) Which type of network (PAN / LAN / MAN / WAN) will be formed when the Durgapur campus is connected to the Delhi head office?
- (e) The company wants to make voice calls over the Internet between Durgapur and Delhi. Name the protocol/technology that can be used for this purpose.

\*\*\*\*\* End of Question Paper \*\*\*\*\*